



最优解 $x_1 = 4, x_2 = 4$

$$Z_{\max} = x_1 + 3x_2 = 16$$

2. ① $\begin{cases} x_1 = 0 \\ x_2 = 0 \end{cases}$

基解 (0,0)

④ $\begin{cases} x_2 = 0 \\ x_1 + x_2 = 8 \end{cases}$

$\begin{cases} x_1 = 8 \\ x_2 = 0 \end{cases}$

基解 (8,0)

② $\begin{cases} x_1 = 0 \\ x_1 + x_2 = 8 \end{cases}$

$\begin{cases} x_1 = 0 \\ x_2 = 8 \end{cases}$

基解 (0,8)

⑤ $\begin{cases} x_2 = 0 \\ x_1 + 2x_2 = 12 \end{cases}$

$\begin{cases} x_1 = 12 \\ x_2 = 0 \end{cases}$

基解 (12,0)

③ $\begin{cases} x_1 = 0 \\ x_1 + 2x_2 = 12 \end{cases}$

$\begin{cases} x_1 = 0 \\ x_2 = 6 \end{cases}$

基解 (0,6)

⑥ $\begin{cases} x_1 + x_2 = 8 \\ x_1 + 2x_2 = 12 \end{cases}$

$\begin{cases} x_1 = 4 \\ x_2 = 4 \end{cases}$

基解 (4,4)

基解即为图中各直线交点

3. 基解 (8,0), (12,0), (4,4) 为可行解, 即为基可行解

4. 可行解 (8,0): $Z = 8$

可行解 (12,0): $Z = 12$

可行解 (4,4): $Z = 16$

∴ 基可行解 (4,4) 能使目标函数最大, 与图解法一致.

5. $\max Z = x_1 + 3x_2$
 $s.t. \begin{cases} x_1 + x_2 - x_3 = 8 \\ x_1 + 2x_2 + x_4 = 12 \\ x_1, x_2, x_3, x_4 \geq 0 \end{cases}$

7. a) $\vec{A}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \vec{A}_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$\vec{a}_1^T = (1 \ 1) \quad \vec{a}_2^T = (1 \ 2)$

b) $x_1 \vec{A}_1 + x_2 \vec{A}_2 = 4 \times \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 4 \times \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 8 \\ 12 \end{pmatrix}$

$\begin{pmatrix} \vec{a}_1^T \vec{x} \\ \vec{a}_2^T \vec{x} \end{pmatrix} = \begin{pmatrix} (1 \ 1) \begin{pmatrix} 4 \\ 4 \end{pmatrix} \\ (1 \ 2) \begin{pmatrix} 4 \\ 4 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 8 \\ 12 \end{pmatrix}$

$A \vec{x} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 4 \\ 4 \end{pmatrix} = \begin{pmatrix} 8 \\ 12 \end{pmatrix}$

∴ $x_1 \vec{A}_1 + x_2 \vec{A}_2 = \begin{pmatrix} \vec{a}_1^T \vec{x} \\ \vec{a}_2^T \vec{x} \end{pmatrix} = A \vec{x}$

6. ① $\begin{cases} x_1 = 0 \\ x_2 = 0 \end{cases} \quad x_3 = -8, x_4 = 12$ 基解 (0,0)

② $\begin{cases} x_1 = 0 \\ x_3 = 0 \end{cases} \quad x_2 = 8, x_4 = -4$ 基解 (0,8)

③ $\begin{cases} x_1 = 0 \\ x_4 = 0 \end{cases} \quad x_2 = 6, x_3 = -2$ 基解 (0,6)

④ $\begin{cases} x_2 = 0 \\ x_3 = 0 \end{cases} \quad x_1 = 8, x_4 = 4$ 基解 (8,0)

⑤ $\begin{cases} x_2 = 0 \\ x_4 = 0 \end{cases} \quad x_1 = 12, x_3 = 4$ 基解 (12,0)

⑥ $\begin{cases} x_3 = 0 \\ x_4 = 0 \end{cases} \quad x_1 = 4, x_2 = 4$ 基解 (4,4)

与问题 2 基解一致